

DURHAM 2 N, NH, USAWMO: **726090**Lat: **43.1716N**Lon: **70.9277W**Elev: **36**StdP: **100.89**Time Zone: **-5.00 (NAE)**Period: **02-19**WBAN: **54794****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-15.1	-25.2	0.4	-14.6	-22.6	0.5	-11.9	5.3	-0.3	4.7	-2.2	1.2	N/A	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	30.9	22.3	29.3	21.4	27.8	20.5	23.8	28.9	22.9	27.5	22.0	26.0	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	16.9	26.2	21.4	16.1	25.3	20.6	15.4	24.4	71.1	28.8	67.7	27.6	64.5	26.0	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.7	3.9	3.5	-22.3	33.8	2.8	1.1	-24.3	34.6	-25.9	35.2	-27.5	35.9	-29.5	36.7		
			-23.0	25.3	2.0	0.6	-24.4	25.7	-25.6	26.0	-26.7	26.4	-28.1	26.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-4.5	-3.2	1.0	7.4	12.9	17.7	21.2	20.3	16.4	10.0	4.4	-1.3
	DBStd	9.90	5.86	5.31	5.11	4.54	4.16	3.70	2.78	2.85	3.77	4.16	4.54	4.89
	HDD10.0	1801	449	369	283	105	16	1	0	0	1	52	177	349
	HDD18.3	3794	707	603	538	330	176	56	6	13	81	259	419	607
	CDD10.0	1286	1	0	3	26	106	231	348	319	193	52	7	1
	CDD18.3	239	0	0	0	1	8	36	96	73	23	1	0	0
	CDH23.3	2112	0	0	4	36	131	336	809	588	198	11	0	0
	CDH26.7	566	0	0	1	9	36	91	233	148	48	1	0	0
Wind	WSAvg	1.4	1.5	1.6	1.8	1.9	1.6	1.3	1.0	0.9	0.9	1.2	1.4	1.4
Precipitation	PrecAvg	1032	77	74	79	94	85	81	77	78	84	98	107	97
	PrecMax	1494	199	262	233	291	237	286	200	224	236	354	286	237
	PrecMin	626	12	0	24	14	18	21	15	21	7	5	16	12
	PrecStd	187	44	41	37	53	45	51	42	41	46	70	54	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	14.1	21.3	27.8	30.5	32.1	32.9	32.2	31.2	25.0	19.8	14.1
		MCWB	12.6	9.0	13.6	16.4	19.2	22.3	23.5	23.6	23.1	18.8	14.5	11.8
	2%	DB	9.0	9.8	15.0	22.7	27.0	29.2	30.8	30.1	27.9	21.9	16.7	10.5
		MCWB	6.1	5.6	10.0	15.2	17.7	21.3	22.6	22.2	21.2	17.0	12.4	8.4
	5%	DB	5.8	6.8	11.2	19.0	24.0	27.0	29.3	28.3	25.5	19.1	14.2	8.2
		MCWB	3.3	3.3	6.3	12.9	16.0	19.3	21.7	21.0	20.0	15.8	10.2	6.0
	10%	DB	3.5	4.2	8.2	15.8	21.1	25.1	27.7	26.6	23.4	16.8	11.9	5.7
		MCWB	1.1	1.4	4.4	9.5	15.2	18.0	21.0	20.0	19.0	13.7	8.9	3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	12.0	15.0	18.1	21.5	23.6	25.2	24.5	24.1	20.4	15.8	12.0
		MCDB	13.8	14.2	22.3	21.9	28.3	29.0	31.0	29.8	29.0	22.6	18.5	12.7
	2%	WB	6.8	7.2	11.7	15.8	19.8	22.3	24.1	23.2	22.5	18.0	13.4	9.1
		MCDB	7.7	10.3	15.9	21.2	24.9	27.6	29.3	27.5	26.7	20.2	15.5	10.4
	5%	WB	3.7	4.2	7.1	13.3	18.0	20.9	23.2	22.2	21.4	16.5	11.1	6.9
		MCDB	5.5	7.1	9.9	17.3	22.5	25.0	27.9	26.1	24.8	18.9	13.4	8.2
	10%	WB	1.4	2.4	4.6	10.4	16.3	19.5	22.3	21.3	20.2	14.7	9.0	4.3
		MCDB	3.2	4.8	7.6	14.3	20.3	23.4	26.4	24.6	22.9	16.9	11.4	6.0
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.9	9.8	11.1	10.9	10.7	10.9	11.3	11.4	10.3	9.4	8.3
		MCDBR	11.1	12.9	14.1	17.3	16.1	13.9	13.2	13.3	13.7	13.2	13.0	11.2
		MCWBR	10.1	9.4	9.3	9.6	7.7	6.8	6.1	6.4	7.3	8.2	9.6	9.0
	5% WB	MCDBR	11.2	13.0	13.8	13.7	13.6	12.1	11.4	10.8	11.7	10.9	11.6	10.2
		MCWBR	10.3	10.0	9.8	9.2	7.8	6.6	6.0	5.9	7.0	8.0	9.7	9.4
Clear-Sky Solar Irradiance	taub	0.285	0.296	0.320	0.356	0.393	0.440	0.459	0.426	0.373	0.333	0.312	0.291	
	taud	2.467	2.449	2.468	2.418	2.353	2.253	2.206	2.285	2.416	2.528	2.547	2.534	
	Ebn at Noon	859	908	925	911	880	837	815	831	855	854	823	822	
	Edh at Noon	72	87	97	111	121	135	140	125	102	80	67	63	
All-Sky Solar Radiation	RadAvg	1.71	2.57	3.64	4.60	5.13	5.52	5.81	5.27	4.24	2.74	1.83	1.37	
	RadStd	0.11	0.20	0.23	0.50	0.59	0.49	0.38	0.26	0.26	0.25	0.16	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	+33

Nomenclature: See separate page